Ma Huamin

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EDUCATION

- **GPA: 3.71/4.0 (Top 10%)** **(2024 – Present)**
- **Core Courses**
  Artificial Intelligence Foundation(94), Probability & Statistics (93), Calculus (95), Data Structures & Algorithms (92), Linear Algebra (91), Discrete Mathematics (88), OOP(C++) (88)
- **English Proficiency**
  CET-4: 522 CET-6: 456

HONORS & AWARDS

- National Endeavor Scholarship (2025)
- University First-Class Individual Scholarship (2025)

PROJECTS

- **Stage-wise NWP Integration for Wind Power Forecasting** **2026.1 – 2026.5**
  Principal Investigator|Research paper | Undergraduate Innovation Training Program
  Wind power forecasting is a key technology for improving renewable energy integration and power system stability, where Numerical Weather Prediction (NWP) provides valuable meteorological guidance. To address the limited understanding of effective NWP integration strategies, we proposed a stage-wise framework incorporating NWP through representation learning, latent fusion, and forecast correction, featuring a plug-and-play neural network-based correction module for forecast bias correction. Experiments on real-world wind power datasets achieved up to 43.2% MAE reduction and 59.5% MSE reduction over baseline forecasting models, while providing practical insights into the design of NWP-aware wind power forecasting systems.
- **NIPT-Based Fetal Anomaly Detection & Optimal Screening Timing** **2025.9**
  Programmer | competition| *China Undergraduate Mathematical Contest in Modeling (CUMCM)*
  To address a noninvasive prenatal testing optimization problem in the 2025 CUMCM, 1) ***multiple regression*** was used to model and test the relationship between DNA concentration, gestational age, and BMI; 2) ***K-means clustering*** was employed to optimize the timing through BMI grouping; 3) ***random survival forest*** was utilized to integrate age and height for predicting success rates, optimizing the detection window and handling censored data; 4) ***random forest classification*** was applied to achieve high-dimensional discrimination of trisomy in female fetuses.

SKILLS

- Programming: Python, C/C++
- Deep Learning: MLP, RNN, Transformer, ResNet
- Frameworks&Tools: PyTorch, Git, Linux
- Data Analysis: Numpy, Pandas, Matplotlib